

PROJECT PHASE 6: Production Deployment & Impact Evaluation

Deadline: Week 11

Weight: 15% of the project grade

Penalty: 3 points per delay (max 2 delays)

Objective

This phase represents the final transition from system → real-world solution.

Students must:

- deploy their application in a real or simulated production environment
- ensure end-to-end usability
- analyze impact, risks, and limitations
- critically evaluate the ethical, societal, and business implications

Coding (Full System & Deployment)

You must extend Phase 5 into a complete, usable, deployed system.

1. Full Application (Final Version)

Upgrade your Phase 5 prototype into a complete version:

Minimum requirements:

- stable application (no crashes)
- clear input → processing → output flow
- proper validation (e.g. invalid inputs)
- improved usability (basic UX improvements)

Examples:

- better UI structure
- loading indicators
- clearer outputs (probabilities, explanations, etc.)

◆ 2. Model Integration (Production-ready)

- model must be:
 - saved (e.g. .pkl, .pt, .onnx)
 - loaded efficiently
- inference must be:
 - fast
 - reproducible

Optional (bonus):

- batch predictions
- API endpoint

◆ 3. Deployment

You must deploy the application in at least one environment:

Accepted options:

- Cloud:
 - AWS (EC2, ECS, Lambda)
 - Azure / GCP
- Local deployment:
 - Docker container
- Demo platforms:
 - Streamlit Cloud
 - HuggingFace Spaces

Minimum requirements:

- application is accessible (link or runnable)
- instructions provided (README)

Desktop deployment means **packaging + distribution + clear execution instructions**.

Example of Acceptable desktop deployment options:

Runnable package

They provide:

- app/
model/
requirements.txt
README.md
run_app.py

And the user can run:

```
pip install -r requirements.txt  
python run_app.py
```

◆ 4. Performance & Scalability Considerations

You must discuss (in code comments or README):

- latency (response time)
- memory usage
- scalability limits

Report (Final Section)

This is the final part of the SAME report (as in previous phases)

1. Deployment Description

- where the system is deployed
- technologies used
- how users interact with it

2. System Architecture

- components description
- data flow
- architecture diagram

3. Ethical & Societal Impact

Students must analyze:

- bias (data / model bias)
- fairness issues
- potential misuse
- privacy concerns

Examples:

- biased predictions
- sensitive data usage
- automation risks

4. Business & Practical Impact

- real-world usability
- potential value:
 - cost reduction
 - automation
 - decision support

5. SWOT analysis

Strategic Assessment of the Solution

Must include:

- Strengths
- Limitations
- Opportunities
- Risks

! Must NOT be just a list — must include explanation and examples.

6. Limitations & Future Work

- what does NOT work well
- scalability limits
- generalization limits
- what could be improved

7. Final Conclusions

- overall findings across ALL phases
- key lessons:
 - data
 - modelling
 - deployment

Deliverables

✓ Report

- LaTeX source
- PDF version

✓ Code

- full project code
- application code
- model files

✓ Deployment

- working link OR
- runnable locally (with instructions)

Submission rule

- before laboratory in Week 11
- everything must be committed (NO zip only)

Grading

- **Report: 50%**
- **Implementation (code + deployment): 50%**

Grading Rubric — Code (Deployment & System)

Section	Points	Full points	Partial points	No points
1 . Full application	3	Functional, stable, complete	Works partially	Missing
2. Model integration	1.5	Efficient loading & inference	Basic integration	Missing
3. Deployment	2.5	Successfully deployed & accessible	Local only / unclear	Missing
4 . Performance & scalability	2	Thoughtful analysis	Superficial	Missing
5 . Code clarity	1	Clean, documented	Superficial	Poor
Total	10 points			

Grading Rubric — Report Phase 6

Section	Points	Full points	Partial points	No points
1 . Deployment description	1.5	Clear, concrete	Basic	Missing
2 . System architecture	1.5	Well explained + diagram	Basic	Missing
3 . Ethical impact	2	Deep, critical analysis	Superficial	Missing
4 . Business impact	1.5	Realistic & concrete	Generic	Missing
5. SWOT Analysis	2	Insightful SWOT analysis	List only	Missing
5 . Limitations & future work	1.5	Well written	Basic	Poor
Total	10 points			